

Grade 11 - Term 3

C1–C4 Practice Exercises and Solutions

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Evaluated criteria

- C1** Characterises continuous random variables by distinguishing them from discrete variables and interpreting their probability density function, cumulative distribution function, and support in economic contexts.
- C2** Applies uniform and triangular continuous distributions with constant or piecewise-linear probability density functions to calculate interval probabilities using geometric areas.
- C3** Applies circular-arc probability density functions to calculate interval probabilities using the geometry of circles and circular segments.
- C4** Applies the exponential distribution to constant-rate waiting-time situations and assesses its suitability.

1 Identifying Probability Distributions from Contexts [C1-5]

A variety of Colombian organizations use probability models to represent uncertain quantities and events. In each situation below, a random variable is defined from the information provided. The appropriate probability distribution must be identified from the characteristics of the situation.

Questions/Tasks.

C1 For each context, characterize the random variable and identify the appropriate probability distribution. Briefly justify your choice based on the features described. [5]

Context 1. A payment platform monitors one scheduled electronic transfer. By the end of processing, it is either completed successfully or not completed. The variable X_1 records the outcome to help estimate verification and support needs.

Context 2. An agricultural technology company monitors the daily temperature variation inside a refrigerated storage unit. The variation X_2 ranges from -2°C to 2°C , with all values in this interval considered equally likely. The company uses this model to set an operating margin.

Context 3. A streaming platform sends the same offer to exactly 80 randomly selected eligible users. Each user either subscribes or does not. The variable X_3 counts the number of subscribers to estimate the server capacity needed for the promotion.

Context 4. A Bogotá parking company records the number of vehicle breakdowns during a fixed one-week period. Comparable records show that breakdowns occur independently and without a predictable timetable. The variable X_4 counts the breakdowns to help schedule maintenance staff.

Context 5. An online marketplace studies the delivery time X_5 for a particular type of order. Delivery times range from 20 to 80 minutes, with deliveries around 45 minutes occurring most frequently. They become progressively less common toward either extreme. The company uses this model to estimate delivery times.

Context 6. A quality-control laboratory tests electronic components until the first defective component is found. Components are tested independently, and testing stops after the first defect. The variable X_6 records the number of components tested, including the defective one, to estimate inspection costs.

Context 7. A cloud-computing provider studies storage usage X_7 among small business customers. Usage ranges from 0 to 50 gigabytes and becomes steadily more common toward the upper limit, without reaching a peak before 50 GB. The provider uses this model to plan capacity.

Context 8. A retailer sends a promotional message to exactly 120 customers selected using the same procedure. Each customer either makes a purchase or does not. The variable X_8 counts the number of purchases to estimate the inventory needed for the campaign.

Context 9. A coffee exporter measures the moisture content X_9 of coffee bags under usual storage conditions. Measurements cluster around 11

Context 10. A food-delivery company records the number of customer cancellations during a fixed three-hour dinner period in one delivery zone. Comparable evenings show cancellations occurring independently without a recurring timetable. The variable X_{10} counts the cancellations to estimate customer-service needs.

2 Delivery Time for a Local Grocery Order [C2-4]

A Colombian grocery-delivery company studies the additional time required to deliver orders during a busy evening period. For a particular delivery zone, the added delivery time X can range from 0 to 10 minutes. A delay of any length within this interval is possible, and shorter delays become steadily more common. Historical data indicate that no additional delay is 4 times as common as a 10-minute delay.

Questions/Tasks.

C2 Determine the complete probability density function for the added delivery time. Then determine the probability that the added delivery time exceeds 7 minutes, and the probability that it is between 3 and 7 minutes. [4]

3 Daily Commute Time for a Bogotá Apprentice [C2-4]

An apprentice records the time spent travelling to a training programme each day. Commute times range from 1 to 9 hours. Times become steadily more common from 1 hour up to a most common time of 4 hours, then steadily less common through 9 hours; there are no commute times right at either limit.

Questions/Tasks.

C2 Determine the complete probability density function for the commute time. Then determine the probability that the commute takes more than 6 hours, and the probability that it takes between 3 and 7 hours. [4]

4 Timing of an Unexpected Market Movement [C3-4]

A financial monitoring system records the time of day at which an unexpected movement in an international currency market occurs. The movement is equally likely to occur at any time during a 24-hour period, so no part of the day is more likely than another.

Questions/Tasks.

C3 Determine the complete probability density function for the time at which the market movement occurs. Then determine the probability that the movement occurs between 9 p.m. and 5 a.m. [4]

5 Direction of an Unexpected Sensor Reading [C3-4]

A directional sensor records the direction in which it is pointing. The sensor can point in any direction around a complete circle, and every direction is equally likely. Let A represent the direction of the sensor, measured in radians from a fixed reference marker at 0 radians.

Questions/Tasks.

C3 Determine the complete probability density function for the direction of the sensor. Then determine which is more probable: the sensor pointing within $\pi/12$ radians of the reference marker, or the sensor pointing within 10° of the reference marker. [4]

6 Waiting for a Study-Room Opening [C4-5]

At a university library in Bogotá, study rooms become available at an average rate of one room every 5 minutes.

Questions/Tasks.

C4 Determine the probability that the next study room becomes available more than 7 minutes after the start of the observation. [4]

7 Waiting for a Market Price Alert

[C4-5]

A financial monitoring system detects significant price changes in a commodity market at an average rate of 3 alerts every 20 minutes.

Questions/Tasks.

C4 Determine the probability that the next significant price change is detected within 6 minutes after the start of the observation. [4]

8 Waiting for a Demand Alert

[C4-5]

An economic forecasting system detects significant changes in consumer demand at an average rate of 4 alerts every 30 minutes.

Questions/Tasks.

C4 Determine the probability that the next significant change in consumer demand is detected between 3 and 8 minutes after the observation time. [4]

9 Processing Investment Transactions

[C4-5]

An investment platform processes buy and sell transactions throughout the trading day. The time between consecutive transactions is modeled by an exponential distribution. Based on previous observations, there is a 75% probability that the next transaction is processed within 8 minutes. On average, how many transactions are expected to be processed in a 10-minute interval?

Questions/Tasks.

C5 Determine the average number of transactions expected to be processed in a 10-minute interval. [5]

10 Waiting for a Loan Approval

[C4-5]

A financial institution processes loan applications throughout the day. The processing time for a randomly selected application is modeled by an exponential distribution. Based on previous observations, there is a 30% probability that an application takes more than 12 minutes to process. Find the average number of applications processed per minute.

Questions/Tasks.

C5 Determine the average number of applications processed per minute. [5]

11 Comparing Investment Transaction Rates

[C4-10]

Two investment platforms process buy and sell transactions throughout the trading day. The transactions processed by each platform are modeled by independent Poisson processes.

For Platform A, there is a 75% probability that the next transaction is processed within 8 minutes. For Platform B, there is a 60% probability that the next transaction is processed within 5 minutes.

Assume both platforms begin processing transactions at the same time. On average, how many minutes does it take for one platform to be ahead of the other by 1 transaction?

Questions/Tasks.

C10 Determine the rate parameter λ for each platform and the expected time until one platform is ahead of the other by 1 transaction. **[10]**